STA 2002
Midterm Review

Outline

- **Point Estimation: MLE and MoM**
- **Confidence Interval**

Details regarding Midterm

- **Date & Time: 19:00 - 21:00, November 3, 2024 (Sunday)**

- **Venue: Liwen Hall**
- **Allowed Items:**
  - **One double-sided A4 cheating sheets (handwritten or typed)**
  - **A non-programmable calculator**

Useful Results for Normal and Chi-square

- If $x_1 \sim N(\mu_1, \sigma_1^2), x_2 \sim N(\mu_2, \sigma_2^2)$ are independently normally distributed variables, then

$$y = x_1 + x_2$$

also follows the normal distribution:

$$y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

If X is $N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0,1)$.

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

Sampling distribution of sample mean is normal, when samples are normal

Useful Results for Normal and Chi-square

If the random variable X is $N(\mu, \sigma^2)$, $\sigma^2 > 0$, then the random variable $V = (X - \mu)^2/\sigma^2 = Z^2$ is $\chi^2(1)$.

Facts:

- (connection to normal) $X_1, X_2, \ldots, X_r \sim N(0,1)$, and they are independent, then

$$X_1^2 + X_2^2 + \cdots + X_r^2 \sim \chi^2(r)$$

- (sum of chi-squares) $X \sim \chi^2(r_1), Y \sim \chi^2(r_2)$, and they are independent, then

$$X + Y \sim \chi^2(r_1 + r_2)$$

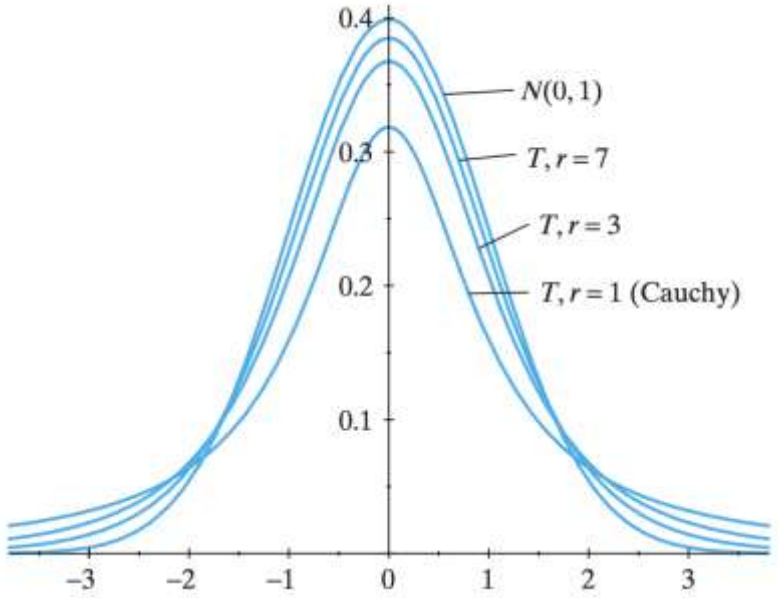
Student's t distribution

- Let $Z \sim N(0,1), U \sim \chi^2(r)$, and they are independent

$$T = \frac{Z}{\sqrt{U/r}} \sim t(r)$$: t distribution with r degrees of freedom

Facts:

The larger the degrees of freedom r, the more the t distribution looks like the normal distribution



Sample mean and variance

- Suppose we have a sequence of i.i.d. (independent and identically distributed) random variables $X_1, X_2, \ldots, X_n$. Let μ denotes their common mean, and σ^2 the common variance

- The **sample mean** is defined as

$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$$\mathbb{E}(\overline{X}) = \mu, \quad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

- The **sample variance** is defined

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$$

Sample mean and variance

If $X_1, X_2, \ldots, X_n$ are observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$, then the distribution of the sample mean $\overline{X} = (1/n)\sum_{i=1}^{n} X_i$ is $N(\mu, \sigma^2/n)$.

Let $X_1, X_2, \ldots, X_n$ be observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$. Then the sample mean,

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

and the sample variance,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \overline{X})^2,$$

are independent and

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{\sigma^2} \text{ is } \chi^2(n-1).$$

Sample mean and variance

- As a consequence, we have

$$U = \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2(n) \qquad W = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{\sigma^2} \sim \chi^2(n-1)$$

$$T = \frac{\overbrace{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}^{N(0,1)}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

$\chi^2(n-1)$

Convergence in Distribution and Probability

Definition 4.1 (Convergence in probability). *Suppose there is a sequence of random variables* $X_n, n = 1, 2, \cdots$ *and a random variable* X. *If for* $\forall \epsilon > 0$, $\exists$ *a constant* $k > 0$ *such that for any* $n \geq k$,

$$\mathbf{P}\left(|X_n - X| > \epsilon\right) \leq \epsilon, \tag{9}$$

then we say X_n *converges to* X *in probability, i.e.,* $X_n \to_p X$.

Definition 4.2 (Convergence in distribution). *Suppose we have a sequence of random variables* $X_n, n = 1, \cdots$ *and a random variable* X. *Suppose for any bounded continuous function* $h(\cdot)$

$$\mathbf{E}[h(X_n)] \to \mathbf{E}[h(X)] \quad \textit{i.e., numerically convergence} \tag{13}$$

then we say that X_n *converges to* X *in distribution, i.e.,* $X_n \to_d X$.

Law of Large Numbers (LLN)

Theorem 5.1. *Suppose random variables* $X_i, i = 1, \cdots$ *are i.i.d. (i.e., independent and identically distributed) with mean* $\mu = \mathbf{E}[X_1]$ *and* $\sigma^2 = \mathrm{Var}(X_1) < \infty$. *Define* $Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, *then*

$$Z_n \to_p \mu. \tag{16}$$

Sample average converges to the population mean.

Central Limit Theorem (CLT)

(Central Limit Theorem) If $\overline{X}$ is the mean of a random sample $X_1, X_2, \ldots, X_n$ of size n from a distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\,\sigma}$$

is $N(0,1)$ in the limit as $n \to \infty$.

Sampling distribution of sample mean is approximately normal, even when samples are NOT normal

Rule of thumb: when n > 30 this works pretty well.

Maximum Likelihood

Method of Moments

POINT ESTIMATOR

Estimator

Suppose X is a random variable with $f(x;\theta)$ as the pdf. If $X_1, X_2, \ldots X_n$ is a random sample of size n from X, the statistic

$$\hat{\Theta} = h(X_1, X_2, \ldots, X_n)$$

Is called a **point estimator** of θ.

After the sample has been selected, $\hat{\Theta}$ takes on a particular numerical value called the **point estimate** of θ.

Note that $\hat{\Theta}$ is a random variable because it is a statistic (function of random variables)

Unbiased Estimator

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

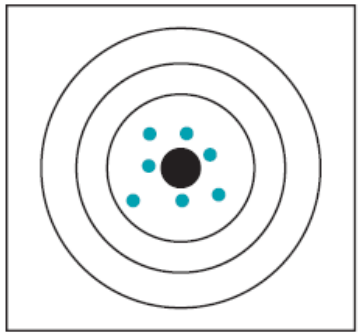
$$E(\hat{\Theta}) = \theta$$

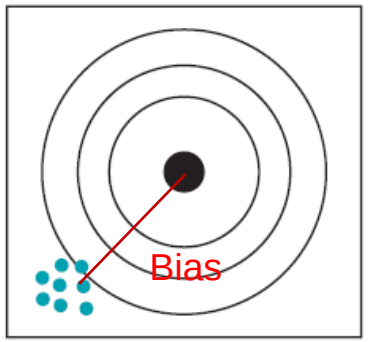
ties to sampling distribution

If the estimator is not unbiased, then the difference

$$E(\hat{\Theta}) - \theta$$

is called the **bias** of the estimator $\hat{\Theta}$.





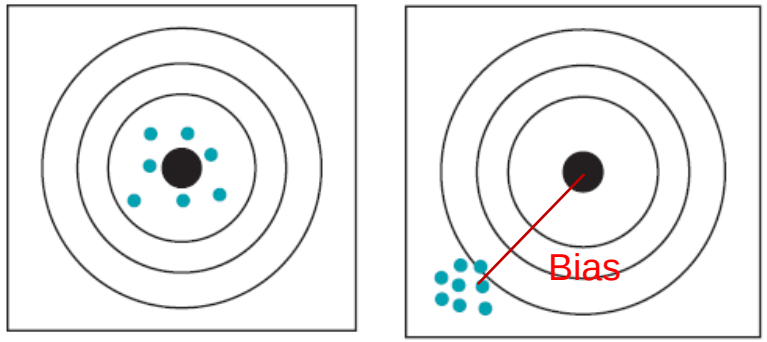
Mean Square Error (MSE)

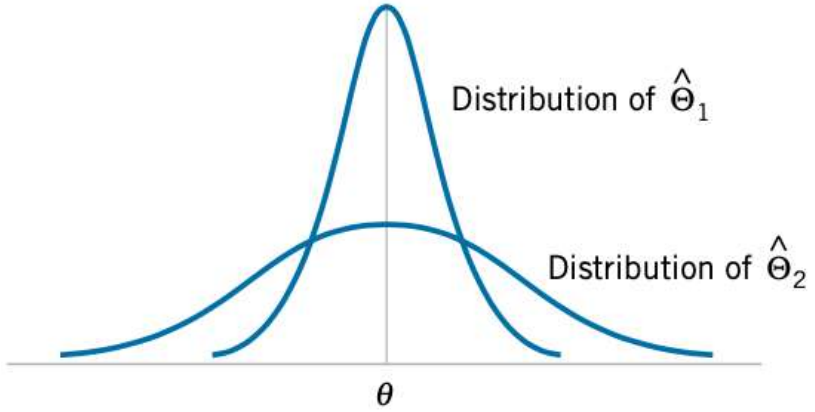
The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$

$$MSE(\hat{\Theta}) = E(\hat{\Theta} - \Theta)^2 = \left[E(\hat{\Theta} - \Theta)\right]^2 + \text{var}(\hat{\Theta} - \Theta)$$

$$MSE(\hat{\Theta}) = \left[\text{Bias}(\hat{\Theta})\right]^2 + \text{var}(\hat{\Theta})$$





Methods for Finding Estimators

- **Assume a distribution for the samples**
- **Estimate the parameter of the distribution**
- **Several methods**
  - **Maximum likelihood**
  - **Method of moment**

Method of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

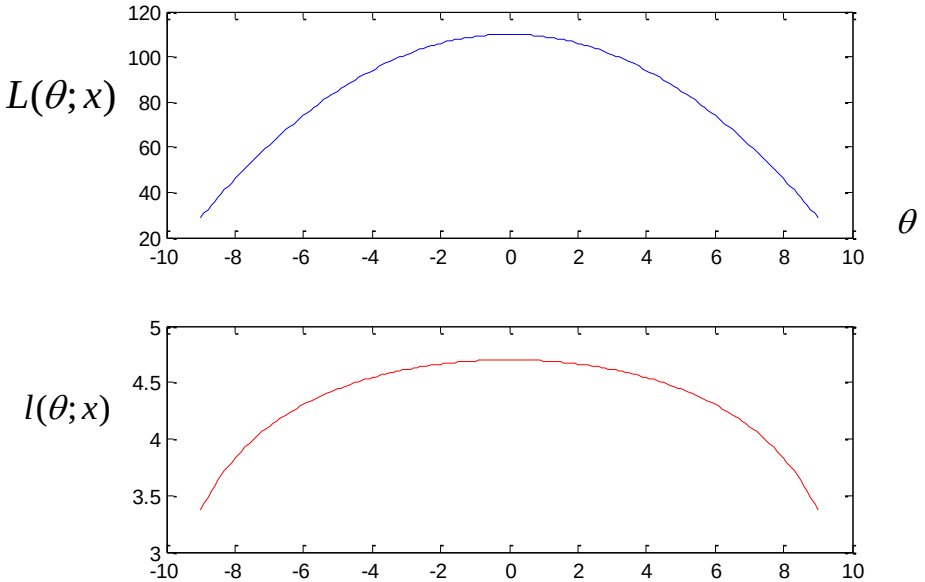
$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n; \theta) \qquad (7\text{-}5)$$

Note that the likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

$$L(\theta; x) = \prod_{i=1}^{n} f(x_i; \theta) = f(x_1; \theta) \ldots f(x_n; \theta)$$

$$l(\theta; x) = \sum_{i=1}^{n} \log\left[f(x_i; \theta) \right]$$

$$\hat{\Theta}(x) = \arg\max_{\theta} L(\theta; x) = \arg\max_{\theta} l(\theta; x)$$



Example: Normal

Let X be normally distributed with unknown μ and known variance σ^2. The likelihood function of a random sample of size n, say $X_1, X_2, \ldots, X_n$, is

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x_i-\mu)^2/(2\sigma^2)} = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-(1/2\sigma^2)\sum_{i=1}^{n}(x_i-\mu)^2}$$

Now

$$\ln L(\mu) = -(n/2)\ln(2\pi\sigma^2) - (2\sigma^2)^{-1}\sum_{i=1}^{n}(x_i-\mu)^2$$

and

$$\frac{d \ln L(\mu)}{d\mu} = (\sigma^2)^{-1}\sum_{i=1}^{n}(x_i-\mu)$$



Check second-order derivative is negative:

$\longrightarrow$ What is the MLE for μ?

$$\ell''(\hat{\mu}) = -\frac{n}{\sigma^2} < 0$$

Example (Continued, unknown variance)

$$\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial (\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0$$

The solutions to the above equation yield the maximum likelihood estimators

$$\hat{\mu} = \overline{X} \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

MLE: Uniform

Let X be a Uniform random variable on the interval $[0, \theta]$

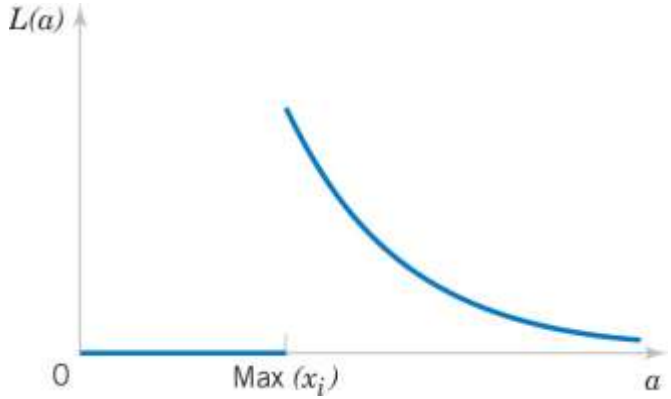
$$f(x;\theta) = \begin{cases} \frac{1}{\theta}, & \text{for } 0 \le x \le \theta, \\ 0, & \text{otherwise,} \end{cases} = \frac{1}{\theta}\mathbf{1}_{\{0 \le x \le \theta\}}$$

indicator function $\mathbf{1}_A(x)$

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{otherwise} \end{cases}$$

The likelihood function of a random sample of size n is:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \frac{1}{\theta^n}\prod_{i=1}^{n}\mathbf{1}_{\{0 \le x_i \le \theta\}} = \begin{cases} \frac{1}{\theta^n}, & \text{if } \theta \ge x_{(n)}, \\ 0, & \text{if } \theta < x_{(n)}. \end{cases}$$



$$\hat{\theta} = X_{(n)} = max\{X_1, X_2, \ldots, X_n\}$$

Calculus methods don't work here because $L(\theta)$ is maximized at the discontinuity. Clearly, θ cannot be smaller than max(x_i), thus the MLE is max(x_i)= $X_{(n)}$.

MLE: Uniform (continued)

Is the MLE of θ an unbiased estimator?



CDF of $X_{(n)}$: for any $x \in (0, \theta)$

$$F(x) = P\left(X_{(n)} \leq x\right) = \prod_{i=1}^{n} P(X_i \leq x) = \left(\frac{x}{\theta}\right)^n$$

PDF of $X_{(n)}$: for any $x \in (0, \theta)$

$$f(x) = F'(x) = \frac{n}{\theta}\left(\frac{x}{\theta}\right)^{n-1}$$

Mean of $X_{(n)}$:

$$E\left[X_{(n)}\right] = \int_0^\theta x f(x) dx = \int_0^\theta \frac{nx}{\theta}\left(\frac{x}{\theta}\right)^{n-1} dx = \frac{n}{\theta^n}\int_0^\theta x^n dx = \frac{n}{n+1}\theta \neq \theta$$

Variance of MLE

Fisher information

Score function $U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$. ← log-likelihood function

Fisher $I_{1}(\theta) = \mathbf{E}\left(\frac{\partial \ell(\theta)}{\partial \theta}\right)^2 = -\mathbf{E}\left[\frac{\partial^2}{\partial \theta^2} \log(f(X_1, \ldots, X_n, \theta))\right]$

$$\sqrt{I_{1}(\theta)}(\widehat{\theta} - \theta) \to N(0, 1),$$

MLE

Equivariance of Maximum likelihood estimators

Theorem 3.1 (Equivariance of MLE). *Let $\tau = g(\theta)$ be a function of θ. Let $\hat{\theta}$ be the MLE of θ. Then, $\hat{\tau} = g(\hat{\theta})$ is the MLE of τ.*

Why MLE is optimal

Theorem 3.1 (Cramér–Rao lower bound). *Suppose θ is an unknown deterministic parameter that is to be estimated from n independent observations (measurements) of X, each from a distribution according to some probability density function $f(x;\theta)$. The variance of any unbiased estimator $\hat{\theta}$ of θ is lower bounded by the reciprocal of the Fisher information:*

$$\operatorname{Var}(\hat{\theta}) \geq \frac{1}{I_{11}(\theta)}.$$

Method of Moments

- **Equating empirical moments to theoretical moments**

Let $X_1, X_2, \ldots, X_n$ be a random sample from either a probability mass function or probability density function with m unknown parameters $\theta_1, \theta_2, \ldots, \theta_m$. The **moment estimators** $\hat{\Theta}_1, \hat{\Theta}_2, \ldots, \hat{\Theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.

***m* equations to estimate *m* parameters**

$$\begin{cases} m_1' = \mu_1' \\ m_2' = \mu_2' \\ \vdots \\ m_m' = \mu_m' \end{cases}$$

Population moments

$$\mu_k' = \begin{cases} \int_x x^k f(x)dx & \text{continuous} \\ \sum_x x^k f(x) & \text{discrete} \end{cases}$$

Sample moments

$$m_k' = \frac{\sum_{i=1}^{n} X_i^k}{n}$$

Algebraic operations of MoM

Theorem 2.2 (CMT for arithmetic operations). *Let X_n, X, Y_n, Y be random variables, and c be some constant. Then*

- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$.*
- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n Y_n \xrightarrow{p} XY$.*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n + Y_n \xrightarrow{d} X + c$. (Slutzky's theorem)*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n Y_n \xrightarrow{d} cX$. (Slutzky's theorem)*

and $g(\cdot)$ is continuous. Then, the method of moments estimator

$$\hat{\theta} = g\left(\frac{1}{n}\sum_{i=1}^{n} X_i, \frac{1}{n}\sum_{i=1}^{n} X_i^2, ..., \frac{1}{n}\sum_{i=1}^{n} X_i^p\right) \xrightarrow{p} \theta_0.$$

Example

MoM estimator for exponential distribution?

$$X \sim Exp\left(\frac{1}{\theta}\right) \qquad f(x) = \frac{1}{\theta} e^{-x/\theta}, \qquad 0 \leq x < \infty$$

Population mean $\mu = \theta$
Sample mean = $\bar{X}$



MoM estimator: $\hat{\theta} = \bar{X}$

Unbiased

Plug-in estimation

Suppose we have a consistent estimator $\hat{\theta}$ for θ, we can *plug in* the estimator $\hat{\theta}$ to the continuous function $g(\cdot)$, and then the estimator for η,

$$\hat{\eta} := g(\hat{\theta}),$$

Two-sided vs. one-sided

CI for mean

CI for mean differences

CI for proportion

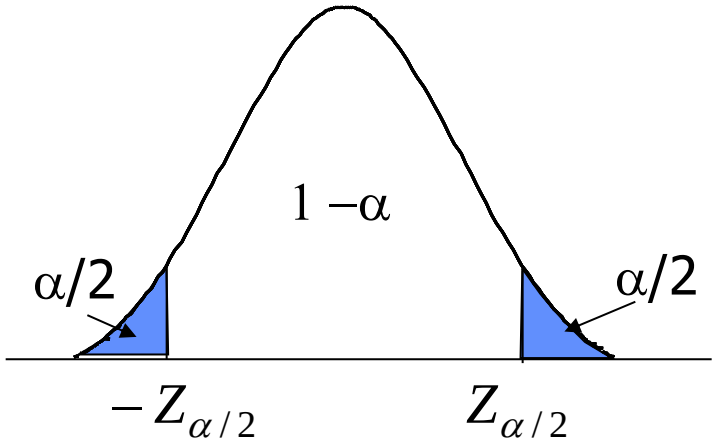
CI for MLE

CONFIDENCE INTERVAL

100(1-α)% CI for Mean of a Normal Distribution (two-sided, known variance)

- **For mean μ from a normal population with known σ,**

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \le \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\} = 1 - \alpha$$

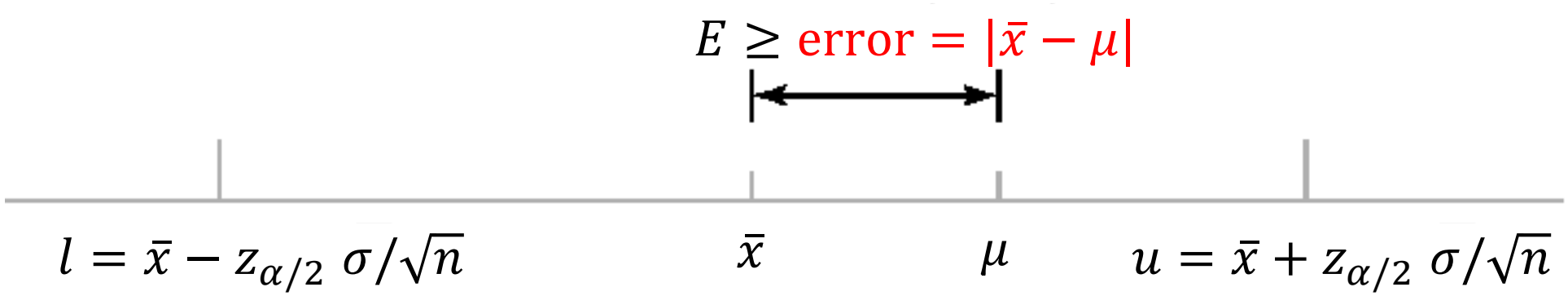
Confidence level

CI Lower CI Upper

$$\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

This CI can be used for mean of non-normal distributions (CLT)

Choice of Sample Size



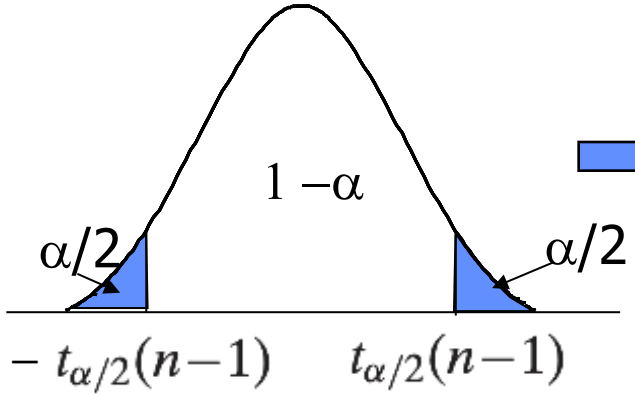
If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

width / length of the CI = $\mathbf{2}\, z_{\alpha/2}\, \sigma/\sqrt{n}$

100(1-α)% CI for Mean of a Normal Distribution (two-sided, unknown variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$



$$P\left(-t_{\alpha/2}(n-1) \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2}(n-1)\right) = 1 - \alpha$$

$$\overline{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \leq \mu \leq \overline{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$$

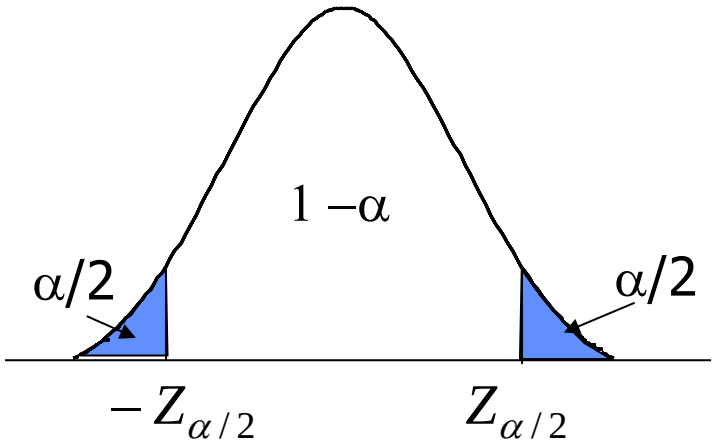
Note that $t_{\alpha/2}(n-1) \approx z_{\alpha/2}$ for *n* >= 30

This CI can be used as approximate CI for mean of non-normal distributions with unknown variance

100(1-α)% CI for a Population Proportion with large sample size

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

- Using CLT, we can estimate a CI for the population proportion.



$$\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n} \sim N(p, \frac{p(1-p)}{n})$$

$$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(Approximate) Two-sided CI

Choice of Sample Size

The sample size for a specified value *E* is given by

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 p(1-p)$$

An upper bound on *n* is given by (i.e., at least (1-α)100% confidence)

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 0.25$$

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If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x}-\mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

Confidence Intervals for Two-Samples

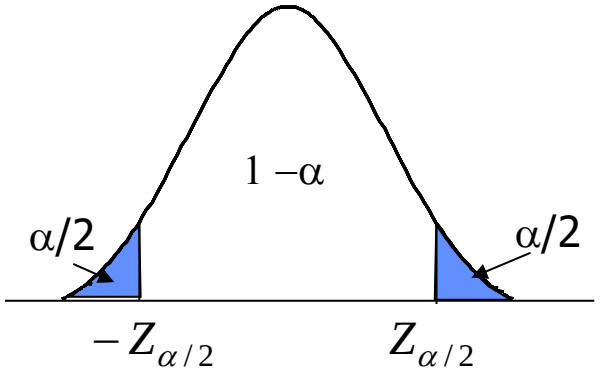
100(1-α)% CI for Differences in Means of Two Normal Distributions (two-sided, known variances)

Two independent random samples from two Normal distributions with the known variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \leq z_{\alpha/2}\right\} = 1 - \alpha$$

← **Confidence level**

CI Lower **CI Upper**

$$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \leq \mu_X - \mu_Y \leq \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$$

This CI can be used for mean of non-normal distributions when *n*,*m*>30

100(1-α)% CI for Difference in Means of Two Normal Distributions (two-sided, unknown but equal variances)

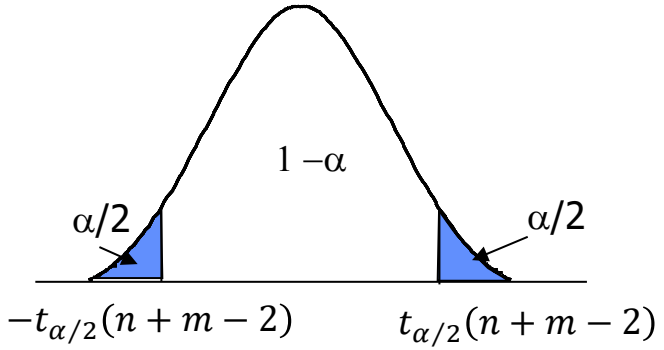
Two independent random samples from two Normal distributions with unknown (but equal) variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$



$$\Rightarrow \quad P\left\{-t_{\alpha/2}(n+m-2) \le \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \le t_{\alpha/2}(n+m-2)\right\} = 1 - \alpha$$

$$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Two-sided

100(1-α)% CI for Difference in Population Proportions with large sample size

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim Bernoulli(p_1) \\ Y_1, Y_2, \cdots, Y_{n_2} \sim Bernoulli(p_2) \end{array}\right\} \rightarrow \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$$

For large sample sizes

Two-sided CI

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \leq (p_1 - p_2) \leq (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

CI for MLE

In order to construct the (two-sided) confidence interval, we first need a consistent estimator for the Fisher information, i.e., find an estimator $\widehat{I}_1(\theta)$ such that

$$\widehat{I}_1(\theta) \to_p I_1(\theta)$$

$$\text{then } \sqrt{n \times \widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta) \to_d N(0,1)$$

Thus $\sqrt{n \times \widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta)$ is a pivot.

To establish the confidence interval, suppose $z_{\alpha/2}$ still being the upper $\alpha/2$ quantile of a standard normal random variable, define the confidence interval

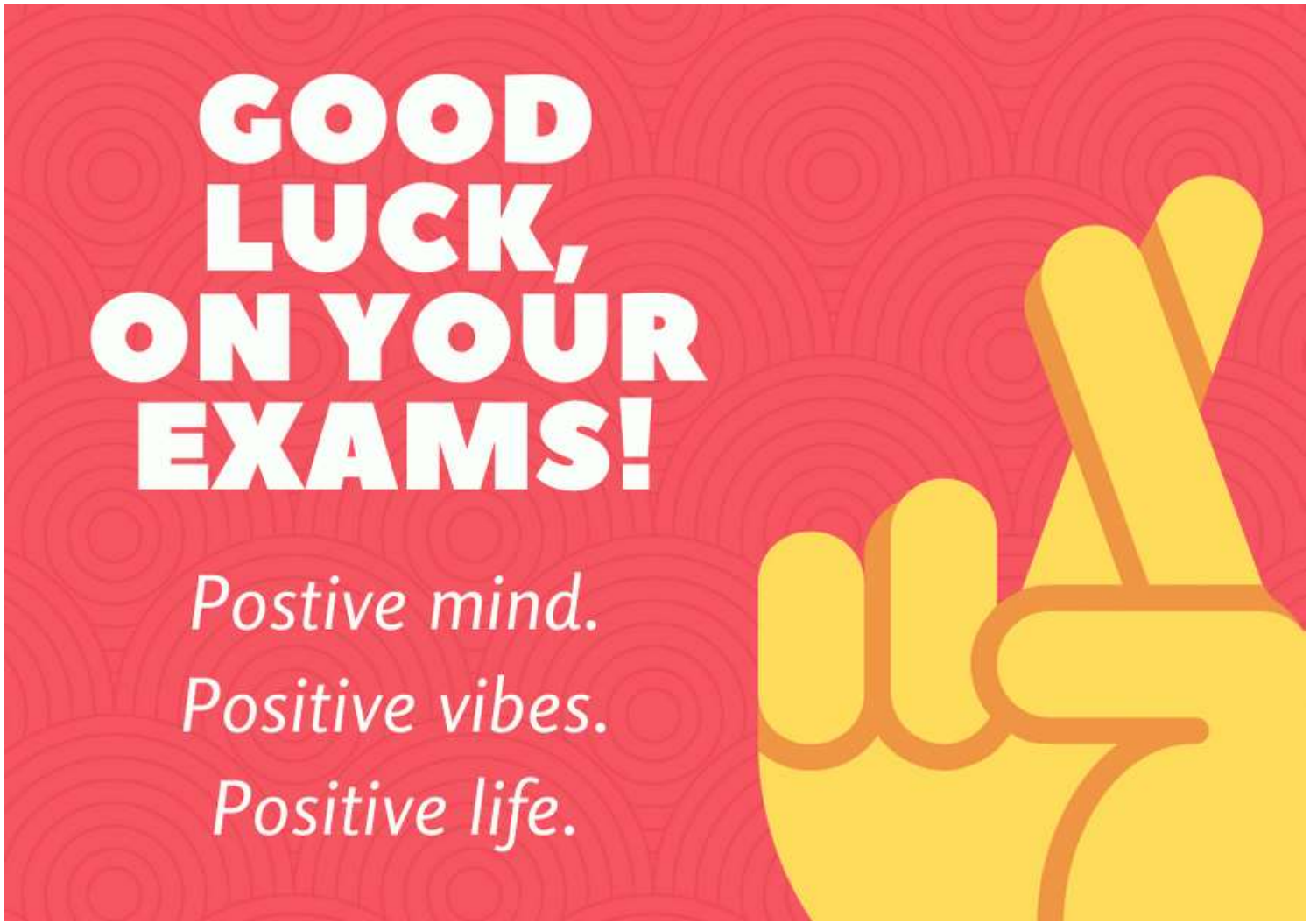
$$\left[\widehat{\theta} - \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}, \quad \widehat{\theta} + \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}\right]$$

Recap:Summary of Sampling Distributions

		One population	Two populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$
	Unknown Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t(n+m-2)$
Sample proportion		$\hat{p} \sim N(p, p(1-p)/n)$	$\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1)$

Summary

		One population	Two populations
Sample Mean	Known Variance	$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \le \mu \le \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$ $\mu \ge \bar{X} - Z_\alpha \times \frac{\sigma}{\sqrt{n}}$ $\quad$ $\mu \le \bar{X} + Z_\alpha \times \frac{\sigma}{\sqrt{n}}$	$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$ $\bar{X} - \bar{Y} - z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y$ $\quad$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
	Unknown Variance	$\bar{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$ $\mu \le \bar{x} + t_\alpha(n-1)(s/\sqrt{n})$ $\mu \ge \bar{x} - t_\alpha(n-1)(s/\sqrt{n})$	$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \ge \bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$
Sample proportion		$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $\hat{p} - Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p$ $\quad$ $p \le \hat{p} + Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ $(\hat{p}_1 - \hat{p}_2) - Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2)$ $\quad$ $(p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
Sample Variance		$\frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}$ $\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}$ $\quad$ $\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_\alpha(n-1)}$	$\frac{S_X^2}{S_Y^2}F_{1-\alpha/2}(m-1,n-1) \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_{\alpha/2}(m-1,n-1)$ $\frac{\sigma_X^2}{\sigma_Y^2} \ge \frac{S_X^2}{S_Y^2}F_{1-\alpha}(m-1,n-1)$ $\quad$ $\frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_\alpha(m-1,n-1)$


GOOD LUCK, ON YOUR EXAMS!
Postive mind.
Positive vibes.
Positive life.